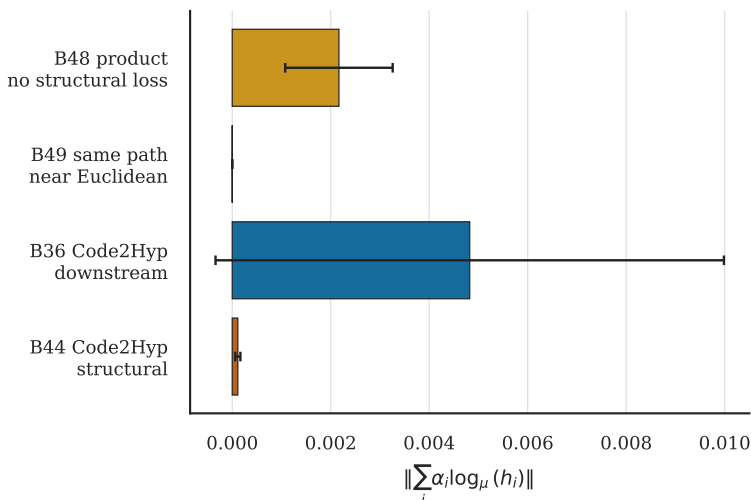
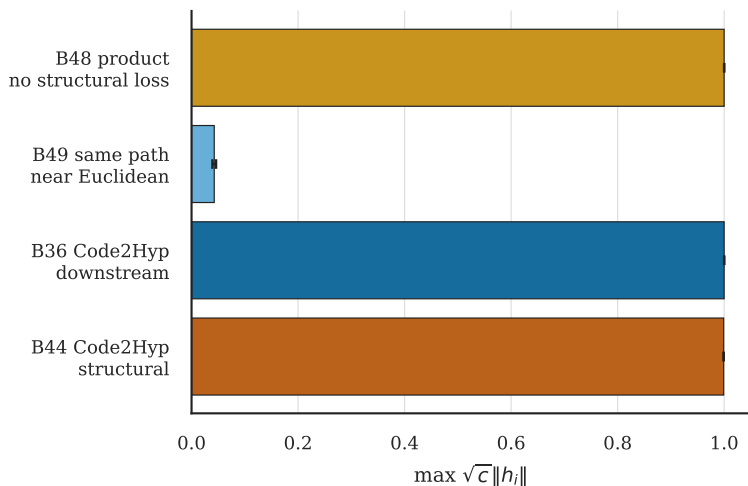


Geometry diagnostics for product-manifold variants

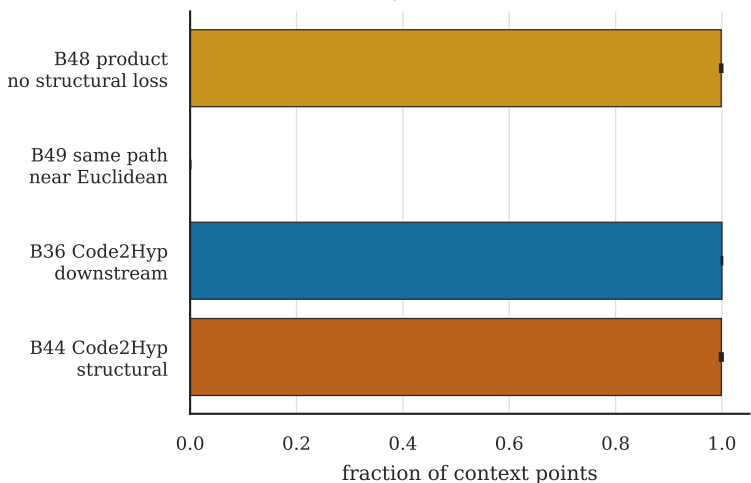
A. Karcher residual



B. Radius utilization



C. Near-boundary rate



D. Final curvature

